

SOURCE REPORT: designs/uart_bt_car/runs/RUN_2026.10.09_09.16.45/reports/signoff/26-mca/rcx_min_sta.checks.rpt

report_checks -unconstrained
Fastest Corner

Startpoint: reset (input port clocked by clk)
Endpoint: _422_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			2.00	2.00	v input external delay
		0.00	0.00	2.00	v reset (in)
1	0.00				reset (net)
		0.00	0.00	2.00	v input2/A (sky130_fd_sc_hd__clkbuf_1)
		0.02	0.05	2.05	v input2/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net2 (net)
		0.02	0.00	2.05	v _373_/A (sky130_fd_sc_hd__buf_4)
		0.06	0.12	2.17	v _373_/X (sky130_fd_sc_hd__buf_4)
8	0.05				_206_ (net)
		0.06	0.00	2.17	v _374_/A (sky130_fd_sc_hd__buf_4)
		0.06	0.14	2.32	v _374_/X (sky130_fd_sc_hd__buf_4)
10	0.06				_207_ (net)
		0.06	0.00	2.32	v _375_/A (sky130_fd_sc_hd__inv_2)
		0.02	0.04	2.36	^ _375_/Y (sky130_fd_sc_hd__inv_2)
1	0.00				_005_ (net)
		0.02	0.00	2.36	^ _422_/SET_B (sky130_fd_sc_hd__dfstp_1)
				2.36	data arrival time
			10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock source latency
		0.05	0.03	10.03	^ clk (in)
1	0.01				clk (net)
		0.05	0.00	10.03	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.05	0.10	10.13	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4	0.04				clknet_0_clk (net)
		0.05	0.00	10.13	^ clkbuf_2_3_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.04	0.09	10.22	^ clkbuf_2_3_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10	0.03				clknet_2_3__leaf_clk (net)
		0.04	0.00	10.22	^ _422_/CLK (sky130_fd_sc_hd__dfstp_1)
			-0.25	9.97	clock uncertainty
			0.00	9.97	clock reconvergence pessimism
			0.09	10.06	library recovery time
				10.06	data required time
				10.06	data required time
				-2.36	data arrival time
				7.70	slack (MET)

Startpoint: _464_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: motor_left_b (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
		0.05	0.04	0.04	^ clk (in)
1	0.01				clk (net)
		0.05	0.00	0.04	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.05	0.11	0.14	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4	0.04				clknet_0_clk (net)
		0.05	0.00	0.14	^ clkbuf_2_2__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.05	0.11	0.25	^ clkbuf_2_2__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
14	0.05				clknet_2_2__leaf_clk (net)
		0.05	0.00	0.25	^ _464_/CLK (sky130_fd_sc_hd__dfrrtp_1)
		0.05	0.27	0.52	v _464_/Q (sky130_fd_sc_hd__dfrrtp_1)
2	0.01				net4 (net)
		0.05	0.00	0.52	v output4/A (sky130_fd_sc_hd__clkbuf_4)
		0.06	0.14	0.66	v output4/X (sky130_fd_sc_hd__clkbuf_4)
1	0.03				motor_left_b (net)
		0.06	0.00	0.66	v motor_left_b (out)
				0.66	data arrival time
			10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock network delay (propagated)
			-0.25	9.75	clock uncertainty

0.00 9.75 clock reconvergence pessimism
-2.00 7.75 output external delay
7.75 data required time

7.75 data required time
-0.66 data arrival time

7.09 slack (MET)

===== Slowest Corner =====

Startpoint: reset (input port clocked by clk)
Endpoint: _422_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description	
			0.00	0.00	clock clk (rise edge)	
			0.00	0.00	clock network delay (propagated)	
			2.00	2.00	v input external delay	
			0.01	2.01	v reset (in)	
1	0.00				reset (net)	
			0.02	2.01	v input2/A (sky130_fd_sc_hd__clkbuf_1)	
			0.05	2.15	v input2/X (sky130_fd_sc_hd__clkbuf_1)	
1	0.00				net2 (net)	
			0.05	2.15	v _373_/A (sky130_fd_sc_hd__buf_4)	
			0.13	2.48	v _373_/X (sky130_fd_sc_hd__buf_4)	
8	0.05				_206_ (net)	
			0.13	2.49	v _374_/A (sky130_fd_sc_hd__buf_4)	
			0.15	2.88	v _374_/X (sky130_fd_sc_hd__buf_4)	
10	0.06				_207_ (net)	
			0.15	2.88	v _375_/A (sky130_fd_sc_hd__inv_2)	
			0.05	2.98	^ _375_/Y (sky130_fd_sc_hd__inv_2)	
1	0.00				_005_ (net)	
			0.05	2.98	^ _422_/SET_B (sky130_fd_sc_hd__dfstp_1)	
				2.98	data arrival time	
				10.00	clock clk (rise edge)	
				0.00	10.00	clock source latency
			0.10	10.07	^ clk (in)	
					clk (net)	
1	0.01				clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)	
			0.10	10.07	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)	
			0.10	10.32	^ clknet_0_clk (net)	
4	0.04				clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)	
			0.10	10.32	^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)	
			0.08	10.55	^ clknet_2_3__leaf_clk (net)	
10	0.03				_422_/CLK (sky130_fd_sc_hd__dfstp_1)	
			0.08	10.56	^	
			-0.25	10.31	clock uncertainty	
			0.00	10.31	clock reconvergence pessimism	
			0.27	10.58	library recovery time	
				10.58	data required time	

				10.58	data required time	
				-2.98	data arrival time	

				7.60	slack (MET)	

Startpoint: _449_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _456_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description
1	0.01	0.10	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
			0.07	0.07	^ clk (in)
			0.10	0.07	^ clk (net)
4	0.04	0.10	0.00	0.07	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.28	0.35	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
			0.10	0.35	^ clknet_0_clk (net)
			0.00	0.35	^ clkbuf_2_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
10	0.03	0.08	0.26	0.62	^ clkbuf_2_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
			0.08	0.62	^ clknet_2_1__leaf_clk (net)
			0.00	0.62	^ _449_/CLK (sky130_fd_sc_hd__dfstp_1)
			0.87	1.48	v _449_/Q (sky130_fd_sc_hd__dfstp_1)
4	0.01	0.16	0.00	1.48	RX.count[12] (net)
			0.00	1.48	v _229_/B (sky130_fd_sc_hd__or4_1)
			1.38	2.86	v _229_/X (sky130_fd_sc_hd__or4_1)
			0.25	2.86	v _103_ (net)
3	0.01	0.25	0.00	2.86	v _232_/A2 (sky130_fd_sc_hd__o41a_1)
			0.99	3.85	v _232_/X (sky130_fd_sc_hd__o41a_1)
			0.20	3.85	v _106_ (net)
			0.00	3.85	v _253_/B (sky130_fd_sc_hd__or4_1)
4	0.01	0.20	0.00	3.85	v _253_/X (sky130_fd_sc_hd__or4_1)
			1.37	5.22	v _123_ (net)
			0.24	5.22	v _260_/A (sky130_fd_sc_hd__or3b_1)
			0.98	6.21	v _260_/X (sky130_fd_sc_hd__or3b_1)

id	type	min	max	value	unit	description
1	0.00	0.16	0.00	6.21 v		_128_ (net)
		0.11	0.75	6.96 v		_261_/S (skyl30_fd_sc_hd_mux2_1)
1	0.00					_261_/X (skyl30_fd_sc_hd_mux2_1)
		0.11	0.00	6.96 v		_129_ (net)
		0.05	0.19	7.15 v		_262_/A (skyl30_fd_sc_hd_clkbuf_1)
1	0.00					_262_/X (skyl30_fd_sc_hd_clkbuf_1)
						062 (net)
		0.05	0.00	7.15 v		_456_/D (skyl30_fd_sc_hd_dfrtp_1)
				7.15		data arrival time
			10.00	10.00		clock clk (rise edge)
			0.00	10.00		clock source latency
		0.10	0.07	10.07 ^		clk (in)
1	0.01					clk (net)
		0.10	0.00	10.07 ^		clkbuf_0_clk/A (skyl30_fd_sc_hd_clkbuf_16)
		0.10	0.25	10.32 ^		clkbuf_0_clk/X (skyl30_fd_sc_hd_clkbuf_16)
4	0.04					clknet_0_clk (net)
		0.10	0.00	10.32 ^		clkbuf_2_3_f_clk/A (skyl30_fd_sc_hd_clkbuf_16)
		0.08	0.23	10.55 ^		clkbuf_2_3_f_clk/X (skyl30_fd_sc_hd_clkbuf_16)
10	0.03					clknet_2_3_leaf_clk (net)
		0.08	0.00	10.55 ^		_456_/CLK (skyl30_fd_sc_hd_dfrtp_1)
			-0.25	10.30		clock uncertainty
			0.03	10.34		clock reconvergence pessimism
			-0.27	10.07		library setup time
				10.07		data required time

				10.07		data required time
				-7.15		data arrival time

				2.93		slack (MET)

		0.07	0.05	0.05	^	clk (in)
1	0.01					clk (net)
		0.07	0.00	0.05	^	clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.16	0.20	^	clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04					clknet_0_clk (net)
		0.06	0.00	0.21	^	clkbuf_2_1_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.05	0.15	0.36	^	clkbuf_2_1_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.03					clknet_2_1_leaf_clk (net)
		0.05	0.00	0.36	^	_449_/CLK (sky130_fd_sc_hd_dfrtp_1)
		0.08	0.42	0.78	v	_449_/Q (sky130_fd_sc_hd_dfrtp_1)
4	0.01					RX.count[12] (net)
		0.08	0.00	0.78	v	_229_/B (sky130_fd_sc_hd_or4_1)
		0.12	0.59	1.37	v	_229_/X (sky130_fd_sc_hd_or4_1)
3	0.01					_103_ (net)
		0.12	0.00	1.37	v	_232_/A2 (sky130_fd_sc_hd_o41a_1)
		0.10	0.50	1.87	v	_232_/X (sky130_fd_sc_hd_o41a_1)
4	0.01					_106_ (net)
		0.10	0.00	1.87	v	_253_/B (sky130_fd_sc_hd_or4_1)
		0.12	0.59	2.46	v	_253_/X (sky130_fd_sc_hd_or4_1)
3	0.01					_123_ (net)
		0.12	0.00	2.46	v	_260_/A (sky130_fd_sc_hd_or3b_1)
		0.07	0.42	2.88	v	_260_/X (sky130_fd_sc_hd_or3b_1)
1	0.00					_128_ (net)
		0.07	0.00	2.88	v	_261_/S (sky130_fd_sc_hd_mux2_1)
		0.05	0.32	3.21	v	_261_/X (sky130_fd_sc_hd_mux2_1)
1	0.00					_129_ (net)
		0.05	0.00	3.21	v	_262_/A (sky130_fd_sc_hd_clkbuf_1)
		0.03	0.10	3.30	v	_262_/X (sky130_fd_sc_hd_clkbuf_1)
1	0.00					_062_ (net)
		0.03	0.00	3.30	v	_456_/D (sky130_fd_sc_hd_dfrtp_1)
				3.30		data arrival time
			10.00	10.00		clock clk (rise edge)
			0.00	10.00		clock source latency
		0.07	0.04	10.04	^	clk (in)
1	0.01					clk (net)
		0.07	0.00	10.04	^	clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.14	10.19	^	clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04					clknet_0_clk (net)
		0.06	0.00	10.19	^	clkbuf_2_3_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.05	0.13	10.32	^	clkbuf_2_3_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.03					clknet_2_3_leaf_clk (net)
		0.05	0.00	10.32	^	_456_/CLK (sky130_fd_sc_hd_dfrtp_1)
		-0.25		10.07		clock uncertainty
		0.02		10.09		clock reconvergence pessimism
		-0.11		9.98		library setup time
				9.98		data required time
				9.98		data required time
				-3.30		data arrival time
				6.68		slack (MET)

```

=====
report_checks --slack_max -0.01
=====

```

```

===== Fastest Corner =====

```

No paths found.

```

===== Slowest Corner =====

```

No paths found.

```

===== Typical Corner =====

```

No paths found.

```

=====
report_check_types -max_slew -max_cap -max_fanout -violators
=====

```

```

===== Fastest Corner =====

```

max fanout

Pin	Limit	Fanout	Slack
clkbuf_2_2_f_clk/X	10	14	-4 (VIOLATED)

```

===== Slowest Corner =====

```

max fanout

Pin	Limit	Fanout	Slack
clkbuf_2_2_f_clk/X	10	14	-4 (VIOLATED)

```

===== Typical Corner =====

```

max fanout

Pin	Limit	Fanout	Slack
clkbuf_2_2__f_clk/X	10	14	-4 (VIOLATED)

=====
report_parasitic_annotation -report_unannotated
=====
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

=====
max slew violation count 0
max fanout violation count 1
max cap violation count 0
=====

=====
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -generated_clocks
=====

=====
SOURCE REPORT: designs/uart_bt_car/runs/RUN_2026.10.09_09.16.45/reports/signoff/26-mca/rcx_min_sta.power.rpt
=====

=====
report_power
=====

=====
Fastest Corner
=====

Group	Internal Power	Switching Power	Leakage Power	Total Power	(Watts)
Sequential	2.67e-04	7.02e-05	2.12e-07	3.37e-04	42.3%
Combinational	2.68e-04	1.92e-04	2.68e-09	4.60e-04	57.7%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	5.35e-04	2.62e-04	2.14e-07	7.98e-04	100.0%
	67.1%	32.9%	0.0%		

=====
Slowest Corner
=====

Group	Internal Power	Switching Power	Leakage Power	Total Power	(Watts)
Sequential	1.79e-04	4.55e-05	8.44e-07	2.25e-04	41.4%
Combinational	1.93e-04	1.24e-04	1.20e-06	3.18e-04	58.6%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.72e-04	1.70e-04	2.05e-06	5.44e-04	100.0%
	68.4%	31.2%	0.4%		

=====
Typical Corner
=====

Group	Internal Power	Switching Power	Leakage Power	Total Power	(Watts)
Sequential	2.31e-04	5.86e-05	2.11e-07	2.89e-04	42.1%
Combinational	2.38e-04	1.61e-04	1.62e-09	3.98e-04	57.9%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	4.68e-04	2.19e-04	2.13e-07	6.88e-04	100.0%
	68.1%	31.9%	0.0%		

=====
SOURCE REPORT: designs/uart_bt_car/runs/RUN_2026.10.09_09.16.45/reports/signoff/26-mca/rcx_min_sta.summary.rpt
=====

=====
report_tns
=====
tns 0.00

=====
report_wns
=====
wns 0.00

=====
report_worst_slack -max (Setup)
=====
worst slack 2.93
=====

report_worst_slack -min (Hold)

worst slack 0.11

SOURCE REPORT: designs/uart_bt_car/runs/RUN_2026.10.09_09.16.45/reports/signoff/28-mca/rcx_max_sta.checks.rpt

report_checks -unconstrained

Fastest Corner

Startpoint: reset (input port clocked by clk)
Endpoint: _422_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description	
1	0.00	0.00	0.00	0.00	clock clk (rise edge)	
			0.00	0.00	clock network delay (propagated)	
			2.00	2.00 v	input external delay	
			0.00	2.00 v	reset (in)	
					reset (net)	
			2.00 v		input2/A (sky130_fd_sc_hd__clkbuf_1)	
			0.02	0.05	2.05 v	input2/X (sky130_fd_sc_hd__clkbuf_1)
						net2 (net)
			0.02	0.00	2.05 v	_373_/A (sky130_fd_sc_hd__buf_4)
			0.06	0.12	2.17 v	_373_/X (sky130_fd_sc_hd__buf_4)
8	0.05			_206_ (net)		
		0.06	0.01	2.18 v	_374_/A (sky130_fd_sc_hd__buf_4)	
10	0.06	0.06	0.15	2.32 v	_374_/X (sky130_fd_sc_hd__buf_4)	
					207 (net)	
1	0.00	0.06	0.00	2.33 v	_375_/A (sky130_fd_sc_hd__inv_2)	
		0.03	0.04	2.37 ^	_375_/Y (sky130_fd_sc_hd__inv_2)	
		0.03	0.00	2.37 ^	_005_ (net)	
				2.37 ^	_422_/SET_B (sky130_fd_sc_hd__dfstp_1)	
1	0.01	0.05			data arrival time	
			10.00	10.00	clock clk (rise edge)	
			0.00	10.00	clock source latency	
			0.04	10.04 ^	clk (in)	
					clk (net)	
			0.05	0.00	10.04 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.05	0.10	10.13 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
						clknet_0_clk (net)
			0.05	0.00	10.14 ^	clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.04	0.09	10.23 ^	clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10	0.03			clknet_2_3__leaf_clk (net)		
		0.04	0.00	10.23 ^	_422_/CLK (sky130_fd_sc_hd__dfstp_1)	
			-0.25	9.98	clock uncertainty	
			0.00	9.98	clock reconvergence pessimism	
			0.09	10.07	library recovery time	
				10.07	data required time	
				10.07	data required time	
				-2.37	data arrival time	
				7.70	slack (MET)	

Startpoint: _464_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: motor_left_b (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
1	0.01	0.05	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
			0.04	0.04 ^	clk (in)
			0.00	0.04 ^	clk (net)
4	0.04	0.05	0.00	0.04 ^	clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.11	0.15 ^	clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
			0.05	0.15 ^	clknet_0_clk (net)
			0.00	0.15 ^	clkbuf_2_2_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
14	0.05	0.05	0.11	0.26 ^	clkbuf_2_2_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
			0.00	0.26 ^	clknet_2_2_leaf_clk (net)
			0.00	0.26 ^	_464_/CLK (sky130_fd_sc_hd__dftrtp_1)
			0.06	0.54 v	_464_/Q (sky130_fd_sc_hd__dftrtp_1)
2	0.01	0.06	0.27	0.54 v	net4 (net)
			0.00	0.54 v	output4/A (sky130_fd_sc_hd__clkbuf_4)
			0.14	0.68 v	output4/X (sky130_fd_sc_hd__clkbuf_4)
			0.00	0.68 v	motor_left_b (net)
1	0.03	0.06	0.00	0.68 v	motor_left_b (out)
			0.00	0.68	data arrival time
			0.00	0.68	
			0.00	0.68	
			10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock network delay (propagated)
			-0.25	9.75	clock uncertainty
			0.00	9.75	

0.00 9.75 clock reconvergence pessimism
-2.00 7.75 output external delay
7.75 data required time

7.75 data required time
-0.68 data arrival time

7.07 slack (MET)

===== Slowest Corner =====

Startpoint: reset (input port clocked by clk)
Endpoint: _422_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description
1	0.00	0.02	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			2.00	2.00	v input external delay
			0.01	2.01	v reset (in)
1	0.00	0.05	0.00	2.01	reset (net)
			0.02	2.01	v input2/A (sky130_fd_sc_hd__clkbuf_1)
8	0.05	0.14	0.00	2.15	v input2/X (sky130_fd_sc_hd__clkbuf_1)
			0.05	2.15	net2 (net)
10	0.06	0.15	0.00	2.15	v _373_/A (sky130_fd_sc_hd__buf_4)
			0.14	2.49	v _373_/X (sky130_fd_sc_hd__buf_4)
1	0.00	0.05	0.01	2.50	_206_ (net)
			0.14	2.50	v _374_/A (sky130_fd_sc_hd__buf_4)
1	0.00	0.05	0.00	2.90	v _374_/X (sky130_fd_sc_hd__buf_4)
			0.05	3.00	^ _375_/A (sky130_fd_sc_hd__inv_2)
1	0.00	0.05	0.11	3.00	^ _375_/Y (sky130_fd_sc_hd__inv_2)
			0.05	3.00	_005_ (net)
1	0.01	0.11	0.00	3.00	^ _422_/SET_B (sky130_fd_sc_hd__dfstp_1)
			0.00	3.00	data arrival time
1	0.01	0.11	10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock source latency
4	0.04	0.10	0.07	10.07	^ clk (in)
			0.11	10.07	clk (net)
10	0.03	0.09	0.00	10.07	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.10	10.33	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
10	0.03	0.09	0.00	10.33	clknet_0_clk (net)
			0.08	10.33	^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
10	0.03	0.09	0.24	10.57	^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
			0.00	10.57	clknet_2_3__leaf_clk (net)
10	0.03	0.09	0.00	10.57	^ _422_/CLK (sky130_fd_sc_hd__dfstp_1)
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	library recovery time
10	0.03	0.09	0.00	10.60	data required time
			-0.25	10.32	clock uncertainty
10	0.03	0.09	0.00	10.32	clock reconvergence pessimism
			0.27	10.60	

Startpoint: _449_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _456_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description
1	0.01	0.11	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
			0.08	0.08	^ clk (in)
					clk (net)
4	0.04	0.10	0.00	0.08	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.28	0.36	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
					clknet_0_clk (net)
10	0.04	0.09	0.00	0.37	^ clkbuf_2_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.27	0.64	^ clkbuf_2_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
					clknet_2_1__leaf_clk (net)
4	0.01	0.16	0.00	0.64	^ _449_/CLK (sky130_fd_sc_hd__dfstp_1)
			0.87	1.51	v _449_/Q (sky130_fd_sc_hd__dfstp_1)
					RX.count[12] (net)
3	0.01	0.26	0.00	1.51	v _229_/B (sky130_fd_sc_hd__or4_1)
			1.39	2.90	v _229_/X (sky130_fd_sc_hd__or4_1)
					103 (net)
4	0.01	0.21	0.00	2.90	v _232_/A2 (sky130_fd_sc_hd__o41a_1)
			1.00	3.91	v _232_/X (sky130_fd_sc_hd__o41a_1)
					106 (net)
3	0.01	0.24	0.00	3.91	v _253_/B (sky130_fd_sc_hd__or4_1)
			1.38	5.29	v _253_/X (sky130_fd_sc_hd__or4_1)
					123 (net)
		0.15	0.00	5.29	v _260_/A (sky130_fd_sc_hd__or3b_1)
			0.99	6.28	v _260_/X (sky130_fd_sc_hd__or3b_1)

1	0.00	0.15	0.00	6.28	v	_128_/S (sky130_fd_sc_hd_mux2_1)
		0.11	0.75	7.03	v	_261_/X (sky130_fd_sc_hd_mux2_1)
1	0.00					_129_ (net)
		0.11	0.00	7.03	v	_262_/A (sky130_fd_sc_hd_clkbuf_1)
		0.06	0.19	7.22	v	_262_/X (sky130_fd_sc_hd_clkbuf_1)
1	0.00					_062_ (net)
		0.06	0.00	7.22	v	_456_/D (sky130_fd_sc_hd_dfrtp_1)
				7.22		data arrival time
			10.00	10.00		clock clk (rise edge)
			0.00	10.00		clock source latency
		0.11	0.07	10.07	^	clk (in)
1	0.01					clk (net)
		0.11	0.00	10.07	^	clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.10	0.26	10.33	^	clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04					clknet_0_clk (net)
		0.10	0.00	10.33	^	clkbuf_2_3_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.08	0.24	10.57	^	clkbuf_2_3_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.03					clknet_2_3_leaf_clk (net)
		0.09	0.00	10.57	^	_456_/CLK (sky130_fd_sc_hd_dfrtp_1)
		-0.25		10.32		clock uncertainty
		0.03		10.36		clock reconvergence pessimism
		-0.27		10.09		library setup time
				10.09		data required time

				10.09		data required time
				-7.22		data arrival time

				2.87		slack (MET)

==== Typical Corner =====

Startpoint: reset (input port clocked by clk)
Endpoint: _422_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			2.00	2.00	v input external delay
		0.01	0.00	2.00	v reset (in)
1	0.00				reset (net)
		0.01	0.00	2.00	v input2/A (sky130_fd_sc_hd_clkbuf_1)
		0.02	0.07	2.08	v input2/X (sky130_fd_sc_hd_clkbuf_1)
1	0.00				net2 (net)
		0.02	0.00	2.08	v _373_/A (sky130_fd_sc_hd_buf_4)
		0.08	0.18	2.26	v _373_/X (sky130_fd_sc_hd_buf_4)
8	0.05				_206_ (net)
		0.08	0.01	2.27	v _374_/A (sky130_fd_sc_hd_buf_4)
		0.08	0.22	2.49	v _374_/X (sky130_fd_sc_hd_buf_4)
10	0.06				_207_ (net)
		0.08	0.00	2.49	v _375_/A (sky130_fd_sc_hd_inv_2)
		0.03	0.06	2.55	^ _375_/Y (sky130_fd_sc_hd_inv_2)
1	0.00				_005_ (net)
		0.03	0.00	2.55	^ _422_/SET_B (sky130_fd_sc_hd_dfstp_1)
				2.55	data arrival time
			10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock source latency
		0.07	0.05	10.05	^ clk (in)
1	0.01				clk (net)
		0.07	0.00	10.05	^ clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.14	10.19	^ clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04				clknet_0_clk (net)
		0.06	0.00	10.19	^ clkbuf_2_3_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.05	0.14	10.33	^ clkbuf_2_3_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.03				clknet_2_3_leaf_clk (net)
		0.05	0.00	10.33	^ _422_/CLK (sky130_fd_sc_hd_dfstp_1)
		-0.25		10.08	clock uncertainty
		0.00		10.08	clock reconvergence pessimism
		0.14		10.22	library recovery time
				10.22	data required time
				10.22	data required time
				-2.55	data arrival time
				7.67	slack (MET)

Startpoint: _449_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _456_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency

		0.07	0.05	0.05	^	clk (in)
1	0.01					clk (net)
		0.07	0.00	0.05	^	clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.16	0.21	^	clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04					clknet_0_clk (net)
		0.06	0.00	0.22	^	clkbuf_2_1_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.15	0.37	^	clkbuf_2_1_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.04					clknet_2_1_leaf_clk (net)
		0.06	0.00	0.37	^	_449_/CLK (sky130_fd_sc_hd_dfrtp_1)
		0.08	0.43	0.79	v	_449_/Q (sky130_fd_sc_hd_dfrtp_1)
4	0.01					RX.count[12] (net)
		0.08	0.00	0.80	v	_229_/B (sky130_fd_sc_hd_or4_1)
		0.13	0.60	1.39	v	_229_/X (sky130_fd_sc_hd_or4_1)
3	0.01					_103_ (net)
		0.13	0.00	1.39	v	_232_/A2 (sky130_fd_sc_hd_o41a_1)
		0.11	0.51	1.90	v	_232_/X (sky130_fd_sc_hd_o41a_1)
4	0.01					_106_ (net)
		0.11	0.00	1.90	v	_253_/B (sky130_fd_sc_hd_or4_1)
		0.12	0.59	2.50	v	_253_/X (sky130_fd_sc_hd_or4_1)
3	0.01					_123_ (net)
		0.12	0.00	2.50	v	_260_/A (sky130_fd_sc_hd_or3b_1)
		0.07	0.42	2.92	v	_260_/X (sky130_fd_sc_hd_or3b_1)
1	0.00					_128_ (net)
		0.07	0.00	2.92	v	_261_/S (sky130_fd_sc_hd_mux2_1)
		0.05	0.33	3.25	v	_261_/X (sky130_fd_sc_hd_mux2_1)
1	0.00					_129_ (net)
		0.05	0.00	3.25	v	_262_/A (sky130_fd_sc_hd_clkbuf_1)
		0.03	0.10	3.35	v	_262_/X (sky130_fd_sc_hd_clkbuf_1)
1	0.00					_062_ (net)
		0.03	0.00	3.35	v	_456_/D (sky130_fd_sc_hd_dfrtp_1)
				3.35		data arrival time
			10.00	10.00		clock clk (rise edge)
			0.00	10.00		clock source latency
1	0.01	0.07	0.05	10.05	^	clk (in)
						clk (net)
		0.07	0.00	10.05	^	clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.14	10.19	^	clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04					clknet_0_clk (net)
		0.06	0.00	10.19	^	clkbuf_2_3_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.05	0.14	10.33	^	clkbuf_2_3_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.03					clknet_2_3_leaf_clk (net)
		0.05	0.00	10.33	^	_456_/CLK (sky130_fd_sc_hd_dfrtp_1)
		-0.25		10.08		clock uncertainty
		0.02		10.10		clock reconvergence pessimism
		-0.11		10.00		library setup time
				10.00		data required time

				10.00		data required time
				-3.35		data arrival time

				6.65		slack (MET)

```

=====
report_checks --slack_max -0.01
=====

```

```

===== Fastest Corner =====

```

No paths found.

```

===== Slowest Corner =====

```

No paths found.

```

===== Typical Corner =====

```

No paths found.

```

=====
report_check_types -max_slew -max_cap -max_fanout -violators
=====

```

```

===== Fastest Corner =====

```

max fanout

Pin	Limit	Fanout	Slack
clkbuf_2_2_f_clk/X	10	14	-4 (VIOLATED)

```

===== Slowest Corner =====

```

max fanout

Pin	Limit	Fanout	Slack
clkbuf_2_2_f_clk/X	10	14	-4 (VIOLATED)

```

===== Typical Corner =====

```

max fanout

Pin	Limit	Fanout	Slack
clkbuf_2_2__f_clk/X	10	14	-4 (VIOLATED)

=====
report_parasitic_annotation -report_unannotated
=====
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

=====
max slew violation count 0
max fanout violation count 1
max cap violation count 0
=====

=====
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -generated_clocks
=====

=====
SOURCE REPORT: designs/uart_bt_car/runs/RUN_2026.10.09_09.16.45/reports/signoff/28-mca/rcx_max_sta.power.rpt
=====

=====
report_power
=====

Fastest Corner					
Group	Internal Power	Switching Power	Leakage Power	Total Power	(Watts)
Sequential	2.66e-04	7.47e-05	2.12e-07	3.41e-04	42.0%
Combinational	2.68e-04	2.03e-04	2.68e-09	4.71e-04	58.0%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	5.35e-04	2.78e-04	2.14e-07	8.12e-04	100.0%
	65.8%	34.2%	0.0%		

=====
Slowest Corner

Group	Internal Power	Switching Power	Leakage Power	Total Power	(Watts)
Sequential	1.78e-04	4.86e-05	8.44e-07	2.28e-04	41.1%
Combinational	1.94e-04	1.31e-04	1.20e-06	3.26e-04	58.9%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.72e-04	1.80e-04	2.05e-06	5.54e-04	100.0%
	67.1%	32.5%	0.4%		

=====
Typical Corner

Group	Internal Power	Switching Power	Leakage Power	Total Power	(Watts)
Sequential	2.30e-04	6.24e-05	2.11e-07	2.93e-04	41.8%
Combinational	2.38e-04	1.70e-04	1.62e-09	4.08e-04	58.2%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	4.68e-04	2.32e-04	2.13e-07	7.00e-04	100.0%
	66.8%	33.2%	0.0%		

=====
SOURCE REPORT: designs/uart_bt_car/runs/RUN_2026.10.09_09.16.45/reports/signoff/28-mca/rcx_max_sta.summary.rpt
=====

=====
report_tns
=====
tns 0.00

=====
report_wns
=====
wns 0.00

=====
report_worst_slack -max (Setup)
=====
worst slack 2.87
=====

report_worst_slack -min (Hold)

worst slack 0.12

SOURCE REPORT: designs/uart_bt_car/runs/RUN_2026.10.09_09.16.45/reports/signoff/30-mca/rcx_nom_sta.checks.rpt

report_checks -unconstrained

Fastest Corner

Startpoint: reset (input port clocked by clk)
Endpoint: _422_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			2.00	2.00	v input external delay
		0.00	0.00	2.00	v reset (in)
1	0.00				reset (net)
		0.00	0.00	2.00	v input2/A (sky130_fd_sc_hd__clkbuf_1)
		0.02	0.05	2.05	v input2/X (sky130_fd_sc_hd__clkbuf_1)
1	0.00				net2 (net)
		0.02	0.00	2.05	v _373_/A (sky130_fd_sc_hd__buf_4)
		0.06	0.12	2.17	v _373_/X (sky130_fd_sc_hd__buf_4)
8	0.05				_206_ (net)
		0.06	0.00	2.18	v _374_/A (sky130_fd_sc_hd__buf_4)
		0.06	0.15	2.32	v _374_/X (sky130_fd_sc_hd__buf_4)
10	0.06				_207_ (net)
		0.06	0.00	2.32	v _375_/A (sky130_fd_sc_hd__inv_2)
		0.03	0.04	2.37	^ _375_/Y (sky130_fd_sc_hd__inv_2)
1	0.00				_005_ (net)
		0.03	0.00	2.37	^ _422_/SET_B (sky130_fd_sc_hd__dfstp_1)
				2.37	data arrival time
			10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock source latency
		0.05	0.03	10.03	^ clk (in)
1	0.01				clk (net)
		0.05	0.00	10.03	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.05	0.10	10.13	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4	0.04				clknet_0_clk (net)
		0.05	0.00	10.13	^ clkbuf_2_3_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.04	0.09	10.23	^ clkbuf_2_3_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10	0.03				clknet_2_3__leaf_clk (net)
		0.04	0.00	10.23	^ _422_/CLK (sky130_fd_sc_hd__dfstp_1)
			-0.25	9.98	clock uncertainty
			0.00	9.98	clock reconvergence pessimism
			0.09	10.06	library recovery time
				10.06	data required time
				10.06	data required time
				-2.37	data arrival time
				7.70	slack (MET)

Startpoint: _464_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: motor_left_b (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout	Cap	Slew	Delay	Time	Description
			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
		0.05	0.04	0.04	^ clk (in)
1	0.01				clk (net)
		0.05	0.00	0.04	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.05	0.11	0.15	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4	0.04				clknet_0_clk (net)
		0.05	0.00	0.15	^ clkbuf_2_2_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
		0.05	0.11	0.26	^ clkbuf_2_2_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
14	0.05				clknet_2_2__leaf_clk (net)
		0.05	0.00	0.26	^ _464_/CLK (sky130_fd_sc_hd__dfrrtp_1)
		0.05	0.27	0.53	v _464_/Q (sky130_fd_sc_hd__dfrrtp_1)
2	0.01				net4 (net)
		0.05	0.00	0.53	v output4/A (sky130_fd_sc_hd__clkbuf_4)
		0.06	0.14	0.67	v output4/X (sky130_fd_sc_hd__clkbuf_4)
1	0.03				motor_left_b (net)
		0.06	0.00	0.67	v motor_left_b (out)
				0.67	data arrival time
			10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock network delay (propagated)
			-0.25	9.75	clock uncertainty

0.00 9.75 clock reconvergence pessimism
-2.00 7.75 output external delay
7.75 data required time

7.75 data required time
-0.67 data arrival time

7.08 slack (MET)

===== Slowest Corner =====

Startpoint: reset (input port clocked by clk)
Endpoint: _422_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description
1	0.00	0.02	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			2.00	2.00	v input external delay
			0.01	2.01	v reset (in)
					reset (net)
1	0.00	0.05	0.02	2.01	v input2/A (sky130_fd_sc_hd__clkbuf_1)
			0.14	2.15	v input2/X (sky130_fd_sc_hd__clkbuf_1)
8	0.05	0.14			net2 (net)
			0.05	2.15	v _373_/A (sky130_fd_sc_hd__buf_4)
			0.34	2.49	v _373_/X (sky130_fd_sc_hd__buf_4)
10	0.06	0.15			_206_ (net)
			0.14	2.49	v _374_/A (sky130_fd_sc_hd__buf_4)
			0.39	2.89	v _374_/X (sky130_fd_sc_hd__buf_4)
1	0.00	0.05			_207_ (net)
			0.15	2.89	v _375_/A (sky130_fd_sc_hd__inv_2)
			0.10	2.99	^ _375_/Y (sky130_fd_sc_hd__inv_2)
		0.05			_005_ (net)
			0.00	2.99	^ _422_/SET_B (sky130_fd_sc_hd__dfstp_1)
			0.00	2.99	data arrival time
1	0.01	0.11	10.00	10.00	clock clk (rise edge)
			0.00	10.00	clock source latency
			0.07	10.07	^ clk (in)
					clk (net)
4	0.04	0.10	0.11	10.07	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.26	10.32	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
10	0.03	0.08			clknet_0_clk (net)
			0.10	10.33	^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.24	10.56	^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
					clknet_2_3__leaf_clk (net)
		0.08	0.00	10.56	^ _422_/CLK (sky130_fd_sc_hd__dfstp_1)
			-0.25	10.31	clock uncertainty
			0.00	10.31	clock reconvergence pessimism
			0.27	10.59	library recovery time
				10.59	data required time
				10.59	data required time
				-2.99	data arrival time
				7.60	slack (MET)

Startpoint: _449_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _456_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout	Cap	Slew	Delay	Time	Description
1	0.01	0.11	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency
			0.08	0.08	^ clk (in)
			0.11	0.08	^ clk (net)
4	0.04	0.10	0.00	0.08	^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.28	0.36	^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
			0.10	0.36	^ clknet_0_clk (net)
10	0.03	0.09	0.00	0.36	^ clkbuf_2_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
			0.27	0.63	^ clkbuf_2_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
			0.09	0.63	^ clknet_2_1__leaf_clk (net)
4	0.01	0.16	0.00	0.63	^ _449_/CLK (sky130_fd_sc_hd__dfstp_1)
			0.87	1.50	v _449_/Q (sky130_fd_sc_hd__dfstp_1)
			0.16	1.50	RX.count[12] (net)
3	0.01	0.26	0.00	1.50	v _229_/B (sky130_fd_sc_hd__or4_1)
			1.39	2.88	v _229_/X (sky130_fd_sc_hd__or4_1)
			0.26	2.88	_103_ (net)
4	0.01	0.20	0.00	2.88	v _232_/A2 (sky130_fd_sc_hd__o41a_1)
			1.00	3.88	v _232_/X (sky130_fd_sc_hd__o41a_1)
			0.20	3.88	_106_ (net)
3	0.01	0.24	0.00	3.88	v _253_/B (sky130_fd_sc_hd__or4_1)
			1.38	5.26	v _253_/X (sky130_fd_sc_hd__or4_1)
			0.24	5.26	_123_ (net)
		0.15	0.00	5.26	v _260_/A (sky130_fd_sc_hd__or3b_1)
			0.98	6.24	v _260_/X (sky130_fd_sc_hd__or3b_1)

1	0.00	0.15	0.00	6.24	v	_128_/S (sky130_fd_sc_hd_mux2_1)
		0.11	0.75	6.99	v	_261_/X (sky130_fd_sc_hd_mux2_1)
1	0.00					_129_ (net)
		0.11	0.00	6.99	v	_262_/A (sky130_fd_sc_hd_clkbuf_1)
		0.05	0.19	7.18	v	_262_/X (sky130_fd_sc_hd_clkbuf_1)
1	0.00					_062_ (net)
		0.05	0.00	7.18	v	_456_/D (sky130_fd_sc_hd_dfrtp_1)
				7.18		data arrival time
		10.00	10.00			clock clk (rise edge)
		0.00	10.00			clock source latency
		0.11	0.07	10.07	^	clk (in)
1	0.01					clk (net)
		0.11	0.00	10.07	^	clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.10	0.26	10.32	^	clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04					clknet_0_clk (net)
		0.10	0.00	10.33	^	clkbuf_2_3_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.08	0.24	10.56	^	clkbuf_2_3_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.03					clknet_2_3_leaf_clk (net)
		0.08	0.00	10.56	^	_456_/CLK (sky130_fd_sc_hd_dfrtp_1)
		-0.25	10.31			clock uncertainty
		0.03	10.35			clock reconvergence pessimism
		-0.27	10.08			library setup time
			10.08			data required time

				10.08		data required time
				-7.18		data arrival time

				2.90		slack (MET)

==== Typical Corner =====

Startpoint: reset (input port clocked by clk)
 Endpoint: _422_ (recovery check against rising-edge clock clk)
 Path Group: asynchronous
 Path Type: max
 Corner: Typical

Fanout	Cap	Slew	Delay	Time	Description

			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (propagated)
			2.00	2.00	v input external delay
		0.01	0.00	2.00	v reset (in)
1	0.00				reset (net)
		0.01	0.00	2.00	v input2/A (sky130_fd_sc_hd_clkbuf_1)
		0.02	0.07	2.08	v input2/X (sky130_fd_sc_hd_clkbuf_1)
1	0.00				net2 (net)
		0.02	0.00	2.08	v _373_/A (sky130_fd_sc_hd_buf_4)
		0.08	0.18	2.26	v _373_/X (sky130_fd_sc_hd_buf_4)
8	0.05				_206_ (net)
		0.08	0.00	2.27	v _374_/A (sky130_fd_sc_hd_buf_4)
		0.08	0.22	2.48	v _374_/X (sky130_fd_sc_hd_buf_4)
10	0.06				_207_ (net)
		0.08	0.00	2.49	v _375_/A (sky130_fd_sc_hd_inv_2)
		0.03	0.06	2.55	^ _375_/Y (sky130_fd_sc_hd_inv_2)
1	0.00				_005_ (net)
		0.03	0.00	2.55	^ _422_/SET_B (sky130_fd_sc_hd_dfstp_1)
				2.55	data arrival time
		10.00	10.00		clock clk (rise edge)
		0.00	10.00		clock source latency
		0.07	0.04	10.04	^ clk (in)
1	0.01				clk (net)
		0.07	0.00	10.04	^ clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.14	10.19	^ clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04				clknet_0_clk (net)
		0.06	0.00	10.19	^ clkbuf_2_3_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.05	0.13	10.32	^ clkbuf_2_3_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.03				clknet_2_3_leaf_clk (net)
		0.05	0.00	10.33	^ _422_/CLK (sky130_fd_sc_hd_dfstp_1)
		-0.25	10.08		clock uncertainty
		0.00	10.08		clock reconvergence pessimism
		0.14	10.22		library recovery time
			10.22		data required time

				10.22	data required time
				-2.55	data arrival time

				7.67	slack (MET)

Startpoint: _449_ (rising edge-triggered flip-flop clocked by clk)
 Endpoint: _456_ (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk
 Path Type: max
 Corner: Typical

Fanout	Cap	Slew	Delay	Time	Description

			0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock source latency

		0.07	0.05	0.05	^	clk (in)
1	0.01					clk (net)
		0.07	0.00	0.05	^	clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.16	0.21	^	clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04					clknet_0_clk (net)
		0.06	0.00	0.21	^	clkbuf_2_1_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.15	0.36	^	clkbuf_2_1_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.03					clknet_2_1_leaf_clk (net)
		0.06	0.00	0.36	^	_449_/CLK (sky130_fd_sc_hd_dfrtp_1)
		0.08	0.42	0.79	v	_449_/Q (sky130_fd_sc_hd_dfrtp_1)
4	0.01					RX.count[12] (net)
		0.08	0.00	0.79	v	_229_/B (sky130_fd_sc_hd_or4_1)
		0.13	0.59	1.38	v	_229_/X (sky130_fd_sc_hd_or4_1)
3	0.01					_103_ (net)
		0.13	0.00	1.38	v	_232_/A2 (sky130_fd_sc_hd_o41a_1)
		0.11	0.51	1.89	v	_232_/X (sky130_fd_sc_hd_o41a_1)
4	0.01					_106_ (net)
		0.11	0.00	1.89	v	_253_/B (sky130_fd_sc_hd_or4_1)
		0.12	0.59	2.48	v	_253_/X (sky130_fd_sc_hd_or4_1)
3	0.01					_123_ (net)
		0.12	0.00	2.48	v	_260_/A (sky130_fd_sc_hd_or3b_1)
		0.07	0.42	2.90	v	_260_/X (sky130_fd_sc_hd_or3b_1)
1	0.00					_128_ (net)
		0.07	0.00	2.90	v	_261_/S (sky130_fd_sc_hd_mux2_1)
		0.05	0.33	3.23	v	_261_/X (sky130_fd_sc_hd_mux2_1)
1	0.00					_129_ (net)
		0.05	0.00	3.23	v	_262_/A (sky130_fd_sc_hd_clkbuf_1)
		0.03	0.10	3.32	v	_262_/X (sky130_fd_sc_hd_clkbuf_1)
1	0.00					_062_ (net)
		0.03	0.00	3.32	v	_456_/D (sky130_fd_sc_hd_dfrtp_1)
				3.32		data arrival time
			10.00	10.00		clock clk (rise edge)
			0.00	10.00		clock source latency
		0.07	0.04	10.04	^	clk (in)
1	0.01					clk (net)
		0.07	0.00	10.04	^	clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.06	0.14	10.19	^	clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4	0.04					clknet_0_clk (net)
		0.06	0.00	10.19	^	clkbuf_2_3_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
		0.05	0.13	10.32	^	clkbuf_2_3_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10	0.03					clknet_2_3_leaf_clk (net)
		0.05	0.00	10.33	^	_456_/CLK (sky130_fd_sc_hd_dfrtp_1)
		-0.25	10.08			clock uncertainty
		0.02	10.09			clock reconvergence pessimism
		-0.11	9.99			library setup time
			9.99			data required time
				9.99		data required time
				-3.32		data arrival time
				6.67		slack (MET)

```

=====
report_checks --slack_max -0.01
=====

```

```

===== Fastest Corner =====

```

No paths found.

```

===== Slowest Corner =====

```

No paths found.

```

===== Typical Corner =====

```

No paths found.

```

=====
report_check_types -max_slew -max_cap -max_fanout -violators
=====

```

```

===== Fastest Corner =====

```

max fanout

Pin	Limit	Fanout	Slack
clkbuf_2_2_f_clk/X	10	14	-4 (VIOLATED)

```

===== Slowest Corner =====

```

max fanout

Pin	Limit	Fanout	Slack
clkbuf_2_2_f_clk/X	10	14	-4 (VIOLATED)

```

===== Typical Corner =====

```

max fanout

Pin	Limit	Fanout	Slack
clkbuf_2_2__f_clk/X	10	14	-4 (VIOLATED)

=====
report_parasitic_annotation -report_unannotated
=====
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

=====
max slew violation count 0
max fanout violation count 1
max cap violation count 0
=====

=====
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -generated_clocks
=====

=====
SOURCE REPORT: designs/uart_bt_car/runs/RUN_2026.10.09_09.16.45/reports/signoff/30-mca/rcx_nom_sta.power.rpt
=====

=====
report_power
=====

Fastest Corner					
Group	Internal Power	Switching Power	Leakage Power	Total Power	(Watts)
Sequential	2.67e-04	7.26e-05	2.12e-07	3.39e-04	42.1%
Combinational	2.68e-04	1.98e-04	2.68e-09	4.66e-04	57.9%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	5.35e-04	2.71e-04	2.14e-07	8.06e-04	100.0%
	66.4%	33.6%	0.0%		

=====
Slowest Corner

Group	Internal Power	Switching Power	Leakage Power	Total Power	(Watts)
Sequential	1.79e-04	4.72e-05	8.44e-07	2.27e-04	41.3%
Combinational	1.93e-04	1.28e-04	1.20e-06	3.23e-04	58.7%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.72e-04	1.75e-04	2.05e-06	5.49e-04	100.0%
	67.7%	31.9%	0.4%		

=====
Typical Corner

Group	Internal Power	Switching Power	Leakage Power	Total Power	(Watts)
Sequential	2.30e-04	6.07e-05	2.11e-07	2.91e-04	41.9%
Combinational	2.38e-04	1.66e-04	1.62e-09	4.03e-04	58.1%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	4.68e-04	2.26e-04	2.13e-07	6.95e-04	100.0%
	67.4%	32.6%	0.0%		

=====
SOURCE REPORT: designs/uart_bt_car/runs/RUN_2026.10.09_09.16.45/reports/signoff/30-mca/rcx_nom_sta.summary.rpt
=====

=====
report_tns
=====
tns 0.00
=====
report_wns
=====
wns 0.00
=====
report_worst_slack -max (Setup)
=====
worst slack 2.90
=====

```
report_worst_slack -min (Hold)
```

```
=====
```

```
worst slack 0.12
```